

Discrete Drainage Dynamics: A Bounded, Conservative Local Rule for Newtonian-Like Potentials

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Abstract

We define a deterministic, parallel, local update rule on a discrete graph. Each node carries a non-negative reserve $R_i \geq 0$, with rest value R_0 at equilibrium. Each oriented link carries a non-negative directional flux. The rule alternates a node update and a link update, with the outgoing fluxes from a node rate-limited so that the reserve cannot become negative. The internal exchange of reserve between nodes is exactly conservative; external sinks are explicit bookkeeping terms. The rule is non-linear only by way of the non-negativity of the reserve, which acts as a finite resource. In its linearised regime, far from empty, the rule's steady state coincides with the discrete Poisson equation, and we measure on the lattice that the static profile around a point source follows a $1/r$ law with the continuum coefficient at the percent level. The contribution of this paper is the rule, not the $1/r$ law: any discrete diffusion in three dimensions yields $1/r$ in the linearised steady state, but the finite-resource, two-step structure of the present rule is what differentiates it from prior lattice models, and is what carries the saturating consequences explored in the companion papers. We are explicit about what we do not derive in this paper.

1 What this paper claims

We make three claims.

1. We define a deterministic, parallel, two-step local rule on a discrete graph. The reserve is bounded below by zero ($R_i \geq 0$, the physical floor: one cannot drain what is not there); R_0 is the rest value, but the reserve is free to exceed it in excited regimes, treated in companion papers. The rule is not the discrete Poisson equation by construction: it is a non-linear update that reduces to the linearised Poisson equation only when the reserve stays well above zero everywhere.
2. Run as a stationary linearised simulation with a point source on a cubic lattice, the rule produces a radial profile compatible with $1/r$ in the bulk, with a coefficient that matches the discrete continuum prediction at the percent level across $L \in \{64, 128, 160\}$.
3. The internal flux-exchange rule preserves total reserve exactly. The non-negativity of the reserve is enforced by rate-limiting the outgoing fluxes and external sinks of a node in proportion, so that emission stops when the reserve reaches zero; no mass is created or destroyed by the internal rule. This is the structural feature that distinguishes the rule from a plain discrete heat equation. (The full rule contains a dormant gauge sector with its own invariants, not used in this paper.)

We do not claim the following. We do not derive Newton’s gravitational constant in physical units; we calibrate it. We do not derive general relativity. We do not derive any property of the Standard Model. We do not show that the rule is unique. We do not assert that the $1/r$ result is itself novel: it is not. What is novel is the rule, and the structural features (reserve non-negativity, rest level R_0 , two-step alternation, rate-limited fluxes) that make the rule non-trivial in the regimes explored in the companion papers.

2 The picture, in plain words

Imagine a fine three-dimensional grid. At each grid point sits a reservoir holding a non-negative amount of stuff. There is a rest level R_0 that the reservoirs all sit at when the system is undisturbed; reservoirs can be excited above R_0 or drained below it, but they cannot go below zero. Between neighbours run pipes that carry a non-negative flow of stuff from the fuller reservoir to the emptier one, at a rate proportional to the level difference. A reservoir cannot deliver more stuff than it actually has: a node at zero stops emitting. There is no separate “maximum flow rate” parameter; the cap follows from what the reservoir contains.

If we start the system at the rest level R_0 everywhere, drain one point, and let the system settle, the bulk reservoir levels relax into a stationary pattern. Far from the source, where every reservoir remains well above zero, the pattern reduces to the textbook discrete Poisson solution and the deficit $R_0 - R_i$ falls as $1/r$ with distance.

The interesting cases are where the floor matters. Near a strong drainage point, the reservoir level can be driven to zero, and emission from that node stops. In this regime the rule is no longer a discrete heat equation: it is a non-linear update with a finite-resource bound. The companion papers explore those regimes (compact-object saturation in Paper II, cosmogenetic regime in Paper III, gauge sector in Papers V–VI). The present paper is restricted to the linearised regime, where the rule reproduces $1/r$.

The rest of this paper makes the rule precise.

3 The substrate

3.1 Graph and variables

The substrate is a graph $G = (V, E)$ in which every node has at least six neighbours and the connectivity is statistically isotropic at large scale. We do not specify the exact topology; in our simulations we use a regular cubic lattice for clarity, and we have verified that BCC gives the same large-scale spectral dimension.

For the purposes of this paper, each node $i \in V$ carries a non-negative reserve $R_i \geq 0$, with rest value R_0 at equilibrium. Each oriented link $(i, j) \in E$ carries a pair of non-negative directional reserve fluxes $F_{i \rightarrow j} \geq 0$ and $F_{j \rightarrow i} \geq 0$. These are the only variables that enter the linearised gravitational simulation reported here. The non-negativity $R_i \geq 0$ is the physical floor (one cannot drain what is not there) and is the only structural source of non-linearity in the rule. In the static gravitational regime treated below, R stays at or below R_0 everywhere because the only source acts as a sink; in excited regimes (companion papers) R can exceed R_0 .

Variables of the wider rule

The full rule, as used in the companion papers, also assigns to each node two angular quadratures (T_i, I_i) subject to a local coherence bound $T_i^2 + I_i^2 \leq R_i$, and to each oriented link a phase $\psi_{ij} \in [-\pi, \pi)$ with $\psi_{ji} = -\psi_{ij}$. These additional variables drive the gauge sector (Papers V–VI), couple to the reserve through the coherence bound, and do not enter the static linearised gravitational regime treated below. We mention them here for completeness; we do not invoke them in the simulation or in the results of this paper.

3.2 Parameters

The rule has the parameters listed in Table 1. They are inputs.

Parameter	Role	Used in this paper
κ	drainage coefficient	yes
α	flux propagation coefficient	yes
R_0	reserve rest value (equilibrium)	yes
J	link relaxation coupling	no (gauge sector, deferred)

Table 1: Parameters of the local rule. The reserve floor at 0 is not a parameter: it is the physical statement that one cannot drain what is not there. R_0 is the rest level at equilibrium, not an absolute ceiling: the reserve can exceed R_0 in excited regimes treated in the companion papers.

4 The two-step tick

Motivation

The class of rules considered here is motivated by an old line of inquiry on whether physics admits a microscopic deterministic, local, iterative description — a “computational substrate” in the sense of Zuse, Fredkin, ’t Hooft, Wolfram, and Lloyd [1–5]. We do not commit to the strong form of this hypothesis. We adopt it as a working stance: it tells us what shape a candidate substrate rule should have — deterministic, local, parallel, conservative, with bounded transport — and the present paper studies one such rule. The numerical claim of the paper does not depend on the working stance.

The rule

The rule decomposes into two alternating half-steps, which we call *Node Update* and *Link Update*. The choice of two-step alternation is a standard one for local lattice theories: it preserves locality (each update reads only from nearest neighbours) and avoids self-reference within a tick. We do not derive the alternation from a deeper principle; we adopt it as a construction.

4.1 Link Update

During the first half-tick, the node reserves are frozen. For each oriented link (i, j) , in parallel, we compute a desired flux,

$$F_{i \rightarrow j}^{\text{des}} = \alpha(R_i - R_j)_+, \quad (1)$$

with α the propagation coefficient. The two directions are independent. Each node then computes a rate-limiting factor that ensures the reserve stays non-negative on its outgoing side. Define the available local budget and the desired total demand,

$$A_i = R_i + \tau S_i^{\text{in}}, \quad S_i^{\text{in}} = \sum_j F_{j \rightarrow i}^{\text{des}}, \quad D_i = \tau O_i^{\text{des}} + \tau \kappa E_i, \quad O_i^{\text{des}} = \sum_j F_{i \rightarrow j}^{\text{des}}, \quad (2)$$

where the demand D_i counts *both* the desired total outflow and the local sink, since both draw from the same reserve. The rate-limiter is

$$\beta_i = \begin{cases} 1, & D_i = 0, \\ \min(1, A_i/D_i), & D_i > 0, \end{cases} \quad (3)$$

and is applied identically to the outflows and to the sink:

$$F_{i \rightarrow j} = \beta_i F_{i \rightarrow j}^{\text{des}}, \quad E_i^{\text{real}} = \beta_i E_i. \quad (4)$$

When demand exceeds the available budget, $\beta_i < 1$ scales outflows and sink down in the same proportion. When the reserve is comfortably above zero, $\beta_i = 1$ and the rule is linear.

4.2 Node Update

During the second half-tick, the link fluxes are frozen. For each node i , in parallel,

$$R_i \leftarrow R_i + \tau S_i^{\text{in}} - \beta_i \tau O_i^{\text{des}} - \beta_i \tau \kappa E_i. \quad (5)$$

By construction of β_i in Eq. (3), this update keeps R_i non-negative at every tick, without any clipping by hand. The internal exchange of reserve between nodes is exactly conservative; the external sources E_i enter as explicit bookkeeping terms (see Sec. 5).

4.3 Why this is not the discrete heat equation

In the regime where every node has a comfortable available budget, $\beta_i = 1$ everywhere, and the rule reduces to a discrete heat equation. Outside that regime, it does not: a node whose available local budget $A_i = R_i + \tau S_i^{\text{in}}$ falls short of its desired demand D_i has $\beta_i < 1$, and its outflows and sink are scaled down in proportion. A node at zero reserve with no incoming compensation stops emitting altogether. The desired outflow is not realised; the reserve does not become negative; no mass is created or lost. This non-linearity is not present in the standard discrete heat equation. It is the structural feature that the companion papers exploit (Paper II for compact-object saturation, Paper III for cosmogenetic regimes).

In terms of the deficit $\delta_i = R_0 - R_i$, the desired flux reads $F_{i \rightarrow j}^{\text{des}} = \alpha (\delta_j - \delta_i)_+$: the reserve flows from the fuller site toward the emptier one.

In the static gravitational regime treated below, where $R \leq R_0$ everywhere, the bound on the flux satisfies $0 \leq F_{i \rightarrow j} \leq \alpha R_0$. This is a downstream consequence of the initial conditions and source type, not a separate cap imposed on the rule.

5 Conservation

Total reserve, exactly conserved by internal exchange. The actual realised fluxes $F_{i \rightarrow j} = \beta_i F_{i \rightarrow j}^{\text{des}}$ enter the Node Update of j as $+\beta_i F_{i \rightarrow j}^{\text{des}}$ (incoming) and the Node Update of i as $-\beta_i F_{i \rightarrow j}^{\text{des}}$ (outgoing). The two contributions cancel when summed over the whole graph, so $\sum_i R_i$ is exactly conserved by the inter-node flux dynamics. The rate-limiter β_i does not violate conservation: it scales outflows and the local sink in the same proportion, so any reduction of an outgoing flux at i is matched by a reduction of the absorbed energy $E_i^{\text{real}} = \beta_i E_i$, never by a creation or destruction of reserve.

External sinks as bookkeeping. The total reserve in the network changes only by $\sum_i E_i^{\text{real}}$, which counts what is taken out by external sources. The closed system consisting of the network plus the external sink reservoir is conservative; the network alone is open, and the sink terms are explicit accounting of that openness, not a violation of conservation.

Dormant gauge sector. The full rule contains a link-phase ψ_{ij} whose closed-loop circulation modulo 2π is preserved by the topology. This invariant is not active in the gravitational regime studied here.

6 Simulation: from rule to $1/r$

6.1 Derivation of the Poisson limit from the rule

In the linear regime $\beta_i = 1$ everywhere, and the desired fluxes coincide with the realised fluxes. The two directional fluxes on a link satisfy

$$F_{i \rightarrow j} - F_{j \rightarrow i} = \alpha (R_i - R_j)_+ - \alpha (R_j - R_i)_+ = \alpha (R_i - R_j),$$

since exactly one of the two terms is non-zero. The net inflow into node i is therefore

$$S_i = \sum_{j \sim i} (F_{j \rightarrow i} - F_{i \rightarrow j}) = \alpha \sum_{j \sim i} (R_j - R_i) = \alpha \Delta R_i,$$

where $\Delta f(i) = \sum_{j \sim i} f(j) - n_i f(i)$ is the discrete graph Laplacian (with $n_i = 6$ on cubic). Writing the Node Update (5) in stationary regime and dividing by τ ,

$$\alpha \Delta R_i = \kappa E_i. \quad (6)$$

Switching to the deficit $\delta_i = R_0 - R_i$, since R_0 is constant we have $\Delta \delta_i = -\Delta R_i$, and

$$-\alpha \Delta \delta_i = \kappa E_i. \quad (7)$$

For a single point source $E_i = E_0 \delta_{\text{Kr}}(i, i_0)$, this is the discrete Poisson equation

$$-\Delta \delta_i = M \delta_{\text{Kr}}(i, i_0), \quad M \equiv \frac{\kappa E_0}{\alpha}, \quad (8)$$

with source strength M set by the imposed sink magnitude E_0 and the microscopic ratio κ/α . The continuum solution is $\delta(r) = M/(4\pi r)$ since $\Delta(1/r) = -4\pi \delta^{(3)}(r)$ in 3D. This is the $1/r$ profile we now measure on the lattice.

6.2 Numerical check

We use a cubic lattice with absorbing (Dirichlet) boundary conditions, iterate to convergence by Jacobi relaxation, bin the field radially, fit $\delta(r) = A/r + B$ over the bulk shell $3 \leq r \leq 0.3L$, and extract A at three lattice sizes $L \in \{64, 128, 160\}$ (Table 2).

L	A_{fit}	relative bias	statistical precision	N_{iter}
64	0.07878	-1.01%	7.0×10^{-5}	5 000
128	0.07866	-1.16%	5.7×10^{-6}	10 000
160	0.07886	-0.90%	3.7×10^{-6}	30 000
$A_{\text{continuum}} = M/(4\pi)$		0.07958		

Table 2: Newton-fit coefficients on cubic lattices of three sizes, $M = 1$, Dirichlet boundary. The fit reproduces the $1/r$ form in all three cases. The bias on the coefficient is roughly L -independent at the percent level, consistent with the discrete-lattice Madelung correction at short range, not with a boundary effect in $1/L$.

6.3 Robustness: spectral dimension

We compute the spectral dimension $d_s(t)$ of the graph Laplacian on cubic and BCC at $L \in \{32, 64, 128\}$ via the heat-kernel return probability $P(t) = \langle e^{-\lambda t} \rangle$, with $d_s(t) = -2 \text{d} \log P / \text{d} \log t$. The plateau value of d_s converges to 3 as L grows. The residual $d_s - 3$ at the plateau decreases on both substrates and at the same rate (Table 3). We do not observe any plateau at a non-integer dimension. The gravitational sector is therefore Newtonian in the IR limit on this class of substrate, as expected.

6.4 Code and data availability

The Python scripts used to reproduce Table 2 (Newton-fit on cubic Poisson) and Table 3 (spectral-dimension comparison cubic vs. BCC), together with the explicit rate-limited rule simulation (`v7_drainage_rule.py`), are available at <https://github.com/stanislasdewavrin/DDD-Paper-1-Foundations>. The scripts are pure-numpy and do not depend on any external simulation framework.

L	cubic ($d_s - 3$ minimum)	BCC ($d_s - 3$ minimum)
32	0.129	0.107
64	0.054	0.045
128	0.023	0.019

Table 3: Spectral-dimension residual at the plateau, on cubic (6 nearest neighbours) and BCC (8 nearest neighbours), at three lattice sizes. Both substrates converge to $d_s = 3$ in the IR limit; BCC sits slightly closer to 3 at every L .

7 Calibration vs. prediction

Calibration. The mapping between the dimensionless deficit field δ_i and the physical Newtonian potential Φ , and the mapping between the local energy budget E_i and physical mass, are external calibrations: in this paper, δ and E remain dimensionless lattice quantities. The lattice unit of length and the parameter κ are fixed by demanding that the coefficient A_{fit} , when matched to Newton’s law, reproduces the measured G for a chosen test mass. This is one calibration condition for two parameters, and it fixes a one-parameter family of physical realisations.

Prediction. Within that family, the rule predicts that the static potential of an isolated point mass is $1/r$ at distances on which the linearised regime is valid, and that finite-resource departures from $1/r$ appear when the bounds become active.

We separate these two cleanly throughout the companion papers. “DDD agrees with X” means prediction; “DDD is calibrated against Y” means calibration.

8 Model checks and extension-level consequences

The table below collects three different kinds of statement that are sometimes lumped together and that we keep separate. It mixes (i) numerical checks internal to the simulation of this paper, (ii) consequences expected of the rule that are worked out in companion papers, and (iii) physical falsifiers that would apply once the rule is calibrated to physical units. We label which is which.

Consequence	Regime	Type	Failure mode / test
$1/r$ static potential	isolated mass, linear regime	numerical check (this paper)	deviation from $1/r$ above bin-statistical noise in the bulk fit window.
Maximum signal speed c_{eff} tied to αR_0 and lattice spacing	all dynamic regimes	structural consequence	faster-than- c_{eff} propagation in simulation; mismatched c from EM and gravitational wavefronts after calibration in the companion paper.
Departure from $1/r$ when $R_i \rightarrow 0$	near-source, very high deficit	extension-level prediction (Paper II)	absence of any cutoff in simulations, or in observed compact-object phenomenology.
Time-reversal asymmetry of U_{tick}	two-step alternation	structural consequence	exact time-reversal symmetry of any prediction that depends on the alternation order.

Table 4: Model checks and consequences expected of the rule, by type.

9 Relation to existing work

The single-point claim of this paper, that a local rule produces a $1/r$ static potential, is by itself trivial: any diffusive process in three dimensions has the same property, and the discrete graph

Laplacian's Green function is a textbook result. The contribution of the paper is the *rule*, not the result. We position it briefly against neighbouring literatures.

Lattice gauge theory [6]. The two-step alternation of node and link updates mirrors the standard matter-gauge structure of Wilson lattice gauge theories, and the link-phase ψ_{ij} in the wider rule plays the role of a U(1) gauge variable. The distinction here is that the link sector carries a directional reserve flux $F_{i \rightarrow j}$, automatically bounded by the non-negativity of the reserve. The presence of a finite-resource floor on a propagating quantity is not standard in lattice gauge theory and is what the companion papers exploit.

Cellular automata for physics [3,4]. Local deterministic rules on discrete substrates have a long history. Most CA programmes target unification of multiple sectors of physics from a single substrate. The present paper is more modest: we define one rule and one regime, and we do not claim more.

Causal sets and discrete-spacetime gravity [7,8]. These programmes treat the substrate itself as dynamical (the graph or simplicial complex evolves). Here the substrate is fixed; the dynamics live on its node and link variables.

Discrete diffusion and graph Laplacian [9]. The static linear regime of our rule is the discrete Poisson equation, and the $1/r$ result is its direct consequence. We do not claim novelty here. The contribution is the embedding of this linear regime inside a non-linear, bounded, two-step rule that admits non-trivial behaviour outside the linear regime. The numerical result in Sec. 6 should accordingly be read as a validation of the implementation and of the linear regime under the specified initial and boundary conditions, not as a new theorem about graph Laplacians.

Entropic and thermodynamic gravity [10,11]. These derive Newton's law from thermodynamic identities on horizons. The present rule is a microscopic mechanical model, not a thermodynamic derivation. The two are not in tension; they target different levels of description.

Lattice fluid models [12]. Lattice Boltzmann methods are conservative, local, parallel updates of density and momentum fields with caps and floors built in for stability. The structural similarity to the present rule is real, and we use lattice-Boltzmann-style implementation techniques in the simulation. The physical interpretation is different: lattice Boltzmann targets fluid mechanics, the present rule targets a substrate from which gravitational and gauge-like behaviour are intended to emerge.

10 What the model is not, and what it does not do

The rule is Newtonian in its present static linearised form. We do not derive general relativity in this paper, and the companion paper on the weak-field limit (Paper II) is a weak-field limit, not a derivation of the full Einstein equations.

We do not derive Newton's gravitational constant in physical units; we calibrate it. A research programme to derive it is sketched in Paper II.

We do not derive the fine-structure constant, particle masses, or any property of the Standard Model.

We do not specify the exact graph topology beyond "minimum 6 neighbours, statistically isotropic". We have tested cubic and BCC, and they agree at the percent level.

We do not give a microscopic origin for the reserve rest value R_0 , which sets the structural scale of the rule. The floor at zero is fixed by the physical statement that one cannot drain what is not there.

We have not shown that the rule is unique. There may exist other local, conservative, bounded rules that produce a $1/r$ profile in their linearised regime.

11 Open questions

Why six neighbours? Six neighbours is the minimal coordination of the cubic stencil used in our simulation; it is not a fundamental requirement of the rule. More general graphs require separate checks of spectral dimension and isotropy. We have no microscopic argument for why the substrate would prefer any specific coordination.

Why this order, Node then Link? The two orderings are not equivalent: U_{tick} is not invariant under exchange. This is a microscopic breaking of time-reversal. We have no microscopic argument for the choice.

What sets R_0 ? R_0 is the rest reserve level at equilibrium, the only structural scale of the rule. The floor at zero is fixed physically. We do not derive R_0 .

Is the rule unique? See Sec. 10.

What sits below the substrate? The graph is the lowest level of description here. Whether it is reducible to anything more fundamental is beyond the present paper.

12 Outlook

The companion papers explore further consequences of the same local rule. We mark each with our own assessment of how well-supported its claims are.

1. Paper II — weak-field gravitational behaviour, including saturation effects when the reserve approaches zero near compact sources. *Linearised plus saturation; partial.*
2. Paper III — discrete-time response and inertial effects. *Speculative.*
3. Paper IV — predictions and tests of the gravitational sector. *Mixed.*
4. Paper V — the gauge sector (T, I, ψ) and emergent electromagnetism. *Speculative; coefficients uncalibrated.*
5. Further exploratory notes (Papers VI and onward) address coupling, cosmology, quantum phenomenology, particle structure, and condensed-matter analogies. *These are increasingly speculative; we say so where it applies.*

The reader should expect the support for our claims to weaken as the series progresses. The present paper, Paper I, claims only the rule itself and the linearised $1/r$ verification. The rest is what one might do with such a rule.

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